Debojit Nath

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Experience

Rivi August 2023 – Present

ML Engineer *March 2024 – Present*

- Drove a 26% gain in response quality with an LLM-as-judge evaluation pipeline that baselined AI services and pinpointed failure modes
- Authored and shipped agent prompts through a centralized, versioned prompt system serving 7 tenants, 2 channels, and 3 environments
- Generated objective and subjective analytics from 30000+ chats for the admin dashboard using conventional methods and AI, cutting manual analysis effort
- Developed end-to-end monitoring of AI agents using Langfuse and Braintrust, enabling per-tenant cost tracking and reducing debugging time
- Developed RAG microservice to supplement LLM service with appropriate context, improving answer relevance
- Cut inference cost 60% by fine-tuning and self-hosting Mistral-7B and Llama-2-13B to replace GPT-4o in production

ML Engineer, Rivi (Client: Purgo) *March 2025 – August 2025*

- Built Graph AI Agent on a Supervisor architecture that autonomously generates solutions for Jira tickets
- Implemented micro-service for Reflection Agent which improved the code execution success rate by 33%
- Implemented LLM-based Evals to assess generated code quality, surfaced in Langfuse traces to flag low-quality outputs before release

ML Intern, Rivi *August 2023 – March 2024*

- Enhanced hotel recommendation by classifying reviews using deep learning methods, improving recommendation relevance
- Integrated the LLM agent with the chat module, making interactions more natural and lifting user engagement

Locobuzz | *Data Science Intern* August 2023 – October 2023

- Predicted stock prices using tweets, achieving 86% accuracy
- Cut sales report turnaround from 3 hours to under 10 minutes by cleaning and structuring raw records into analysis-ready tables

Shortlyst AI | *Data Scientist* November 2022 – July 2023

- Developed LinkedIn data scraping bots and stored 30M+ profiles in MongoDB
- Fine-tuned DistilBERT model to predict industry tags for companies lacking them
- Automated stats generation of bot performance for monitoring via Slack, saving hours of manual reporting

Projects

Toxic Comment Classification | *NLP, Multi-label Classification, Kaggle*

- Trained a multi-label classifier for the Kaggle Toxic Comment Classification Challenge, scoring 0.9845 on the public leaderboard

Fashion Apparel Classification | *PyTorch, Vision Transformer*

- Trained a Vision Transformer on an imbalanced apparel dataset, applying re-sampling to counter class imbalance and lifting accuracy from 66% to 69%

Technical Skills

Languages: Python, SQL, C++, Bash
GenAI / LLM: LangChain, LangGraph, RAG, Vector Databases, Agentic Workflows, Multi-Agent Systems, Prompt Engineering, Tool / Function Calling, LLM-as-Judge Evals, LoRA / PEFT Fine-tuning, vLLM, OpenAI & Anthropic APIs
LLMOps: Langfuse, Braintrust, Prompt Versioning & Rollout, Tracing, Token & Cost Tracking
ML & NLP: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, Vision Transformers, Recommender Systems, Sentiment Analysis, OpenCV
Data & Infra: pandas, NumPy, PostgreSQL, MongoDB, Redis, FastAPI, Flask, REST APIs, Microservices, Docker, Kubernetes, AWS, Git, CI/CD, Selenium, BeautifulSoup

Education

Indian Institute of Technology Madras | *Diploma in Data Science* 2021 – 2023

Techno India Institute of Technology | *Bachelor of Computer Application, CGPA: 8.5* 2018 – 2021